

Pathology-Level ML Model Performance Summary (Test Set) Classical ML models trained on neural network pathology predictions							
Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Attention Pool	A Lines	Random Forest	0.686	[0.609-0.763]	41632	0.471	5
Attention Pool	A Lines	Logistic Regression	0.742	[0.674-0.809]	41632	0.471	5
Attention Pool	A Lines	XGBoost	0.729	[0.657-0.802]	41632	0.471	5
Attention Pool	Large Consolidation	Random Forest	0.778	[0.669-0.887]	41632	0.111	5
Attention Pool	Large Consolidation	Logistic Regression	0.875	[0.820-0.929]	41632	0.111	5
Attention Pool	Large Consolidation	XGBoost	0.821	[0.722-0.920]	41632	0.111	5
Attention Pool	Pleural Effusion	Random Forest	0.667	[0.627-0.706]	41632	0.011	5
Attention Pool	Pleural Effusion	Logistic Regression	0.744	[0.718-0.771]	41632	0.011	5
Attention Pool	Pleural Effusion	XGBoost	0.834	[0.801-0.868]	41632	0.011	5
Attention Pool	Other Pathology	Random Forest	0.589	[0.526-0.652]	41632	0.404	5
Attention Pool	Other Pathology	Logistic Regression	0.630	[0.614-0.645]	41632	0.404	5
Attention Pool	Other Pathology	XGBoost	0.622	[0.562-0.683]	41632	0.404	5
Original Noinitweights	A Lines	Random Forest	0.702	[0.610-0.793]	43689	0.472	5
Original Noinitweights	A Lines	Logistic Regression	0.753	[0.688-0.817]	43689	0.472	5
Original Noinitweights	A Lines	XGBoost	0.737	[0.651-0.823]	43689	0.472	5
Original Noinitweights	Large Consolidation	Random Forest	0.747	[0.635-0.858]	43689	0.110	5
Original Noinitweights	Large Consolidation	Logistic Regression	0.867	[0.826-0.908]	43689	0.110	5
Original Noinitweights	Large Consolidation	XGBoost	0.806	[0.730-0.882]	43689	0.110	5
Original Noinitweights	Pleural Effusion	Random Forest	0.613	[0.552-0.674]	43689	0.011	5
Original Noinitweights	Pleural Effusion	Logistic Regression	0.843	[0.782-0.903]	43689	0.011	5
Original Noinitweights	Pleural Effusion	XGBoost	0.785	[0.728-0.842]	43689	0.011	5
Original Noinitweights	Other Pathology	Random Forest	0.601	[0.561-0.640]	43689	0.405	5
Original Noinitweights	Other Pathology	Logistic Regression	0.653	[0.609-0.697]	43689	0.405	5
Original Noinitweights	Other Pathology	XGBoost	0.640	[0.604-0.676]	43689	0.405	5
Attention Pool Noinitweights	A Lines	Random Forest	0.715	[0.672-0.758]	43689	0.472	5
Attention Pool Noinitweights	A Lines	Logistic Regression	0.773	[0.739-0.806]	43689	0.472	5
Attention Pool Noinitweights	A Lines	XGBoost	0.760	[0.719-0.801]	43689	0.472	5
Attention Pool Noinitweights	Large Consolidation	Random Forest	0.838	[0.793-0.884]	43689	0.110	5
Attention Pool Noinitweights	Large Consolidation	Logistic Regression	0.895	[0.878-0.912]	43689	0.110	5
Attention Pool Noinitweights	Large Consolidation	XGBoost	0.880	[0.833-0.927]	43689	0.110	5
Attention Pool Noinitweights	Pleural Effusion	Random Forest	0.633	[0.533-0.733]	43689	0.011	5
Attention Pool Noinitweights	Pleural Effusion	Logistic Regression	0.829	[0.768-0.891]	43689	0.011	5
Attention Pool Noinitweights	Pleural Effusion	XGBoost	0.795	[0.718-0.871]	43689	0.011	5
Attention Pool Noinitweights	Other Pathology	Random Forest	0.610	[0.588-0.631]	43689	0.405	5
Attention Pool Noinitweights	Other Pathology	Logistic Regression	0.684	[0.641-0.728]	43689	0.405	5
Attention Pool Noinitweights	Other Pathology	XGBoost	0.658	[0.613-0.703]	43689	0.405	5